

OPEN TRACK
EduTech

DevDragons

PEC HACKS
4.0

EduDNA

THE AI LEARNING GENOME



Problem Statement

≈

Marks & attendance are not learning. Current platforms score outcomes but never see HOW a student actually learns, retains, solves problems, or collaborates.

=

One experience for every learner. Because cognitive profiles are never captured, all students are pushed through the same pace, content, and format regardless of how they think.

∅

No memory of the learning journey. Retention, curiosity, and problem-solving patterns are never tracked over time — every term starts from a blank slate.

SOLUTION



Continuous Behavioral Capture

Streams learning activity, quiz attempts, retries, and collaboration signals as the student works.



Knowledge Graph Genome

Builds a living graph linking concepts, mistakes, retention and curiosity for every learner.



Reinforcement-Learned Pathing

An RL agent recommends the next best concept, pace, and format to reinforce retention.



Problem-Solving Pattern Mining

Detects how a student approaches problems — strategy, persistence, and collaboration style.



Generative AI Insight Layer

Turns the raw genome into plain-language insights for teachers, parents, and the student.



KEY OUTCOMES

An AI genome that turns every learner into a continuously understood, uniquely supported individual.

- Personalized pace, content & format per learner
- Early prediction of learning success or struggle
- Genome evolves with the student, term after term
- Actionable insight for teachers, not just scores

WHERE EXISTING PLATFORMS FAIL

Static grades

One score, no story

Siloed data

LMS, quizzes never linked

No behavior lens

Curiosity, grit unseen

Fixed content

Same path for all

1

Lifelong genome, not a static grade.

Every learner carries one continuously-built EduDNA profile instead of a fresh, forgettable scoresheet each term.

2

Predicts future learning success.

Behavioral patterns — retention curves, curiosity signals, persistence — forecast outcomes before a test ever happens.

3

Evolves as the student learns.

The genome is never final; graph neural networks update it with every interaction, keeping it current.

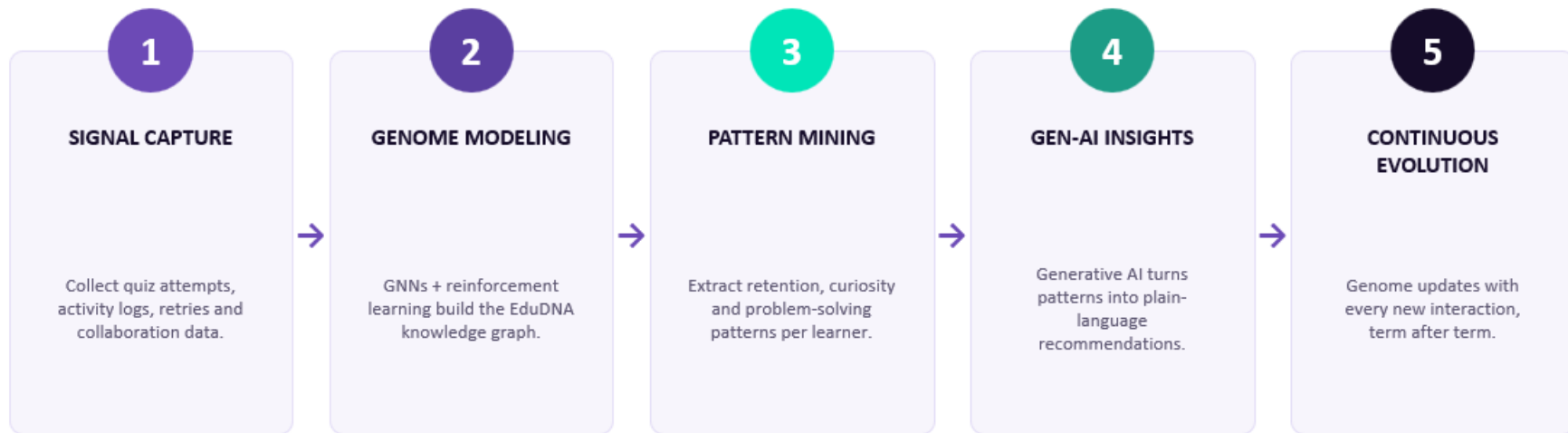
4

Sees the whole learner, not just answers.

Captures memory retention, curiosity, collaboration and problem-solving depth — dimensions grades never show.

EduDNA — Development Workflow

From raw learning signals to a living, evolving genome



GOAL: Give every learner a living digital genome that predicts and personalizes their path to success.

Dependencies and Showstopper

1

Rich, continuous learning data

Accurate genomes need quiz attempts, activity logs, and collaboration signals — sparse data weakens the model.

2

LMS & school system integration

Requires scalable knowledge-graph infrastructure and clean integration with existing LMS/ERP platforms.

3

Data quality risk

Incomplete or inconsistent learning records can distort retention scores and produce misleading recommendations.

4

Privacy & ethical safeguards.

Behavioral profiling of students demands strict consent, anonymization, and responsible-AI guardrails.

Additional Info



EduDNA replaces static report cards with a living AI-built genome for every learner. By continuously modeling retention, curiosity, collaboration and problem-solving through knowledge graphs and reinforcement learning, it predicts learning success early and personalizes each student's path — giving teachers a genuine understanding of how, not just how well, each student learns.

AI Features

- Learning Genome Profiling
- Retention Prediction
- Curiosity Scoring
- Collaboration Analysis
- Problem-Solving Pattern Mining
- Success Prediction

Technologies Used

- Generative AI
- Knowledge Graphs
- Reinforcement Learning
- Graph Neural Networks
- Neo4j / PostgreSQL
- React.js / Flutter

Target Users

- Schools & Universities
- EdTech Platforms
- Teachers & Mentors
- Students & Parents
- Corporate L&D Teams

Future Enhancements

- Adaptive curriculum generation
- Genome-based mentor matching
- Multi-modal genome (voice/video)
- Cross-institution portability
- Explainable AI for teachers

Frontend Framework

- React.js — fast, interactive web dashboard
- Flutter — cross-platform mobile app

Backend & API

- Node.js / FastAPI — services & REST APIs
- PostgreSQL — student & activity records

Genome Engine

- Neo4j — knowledge-graph EduDNA store
- Graph Neural Networks — pattern learning

TECH STACK

AI & Machine Learning

- Reinforcement Learning — path recommendation
- Generative AI — plain-language insights

Infrastructure & Cloud

- AWS / Azure — scalable hosting
- Docker & GitHub — deployment & collaboration

UI & Styling

- Tailwind CSS — modern responsive UI
- Figma — UX prototypes & wireframes

Team Name: DevDragons

Koppala Naveen	Saveetha Engineering College	Chennai	Frontend Developer	Male
S L Narasimha Reddy	Saveetha Engineering College	Chennai	Backend Developer	Male
K Madhava Reddy	Saveetha Engineering College	Chennai	AI & ML Engineer	Male
K Abhineswar Reddy	Saveetha Engineering College	Chennai	Data & Knowledge Graph Engineer	Male
P Aditya Naga Sai	Saveetha Engineering College	Chennai	UI/UX Designer	Male